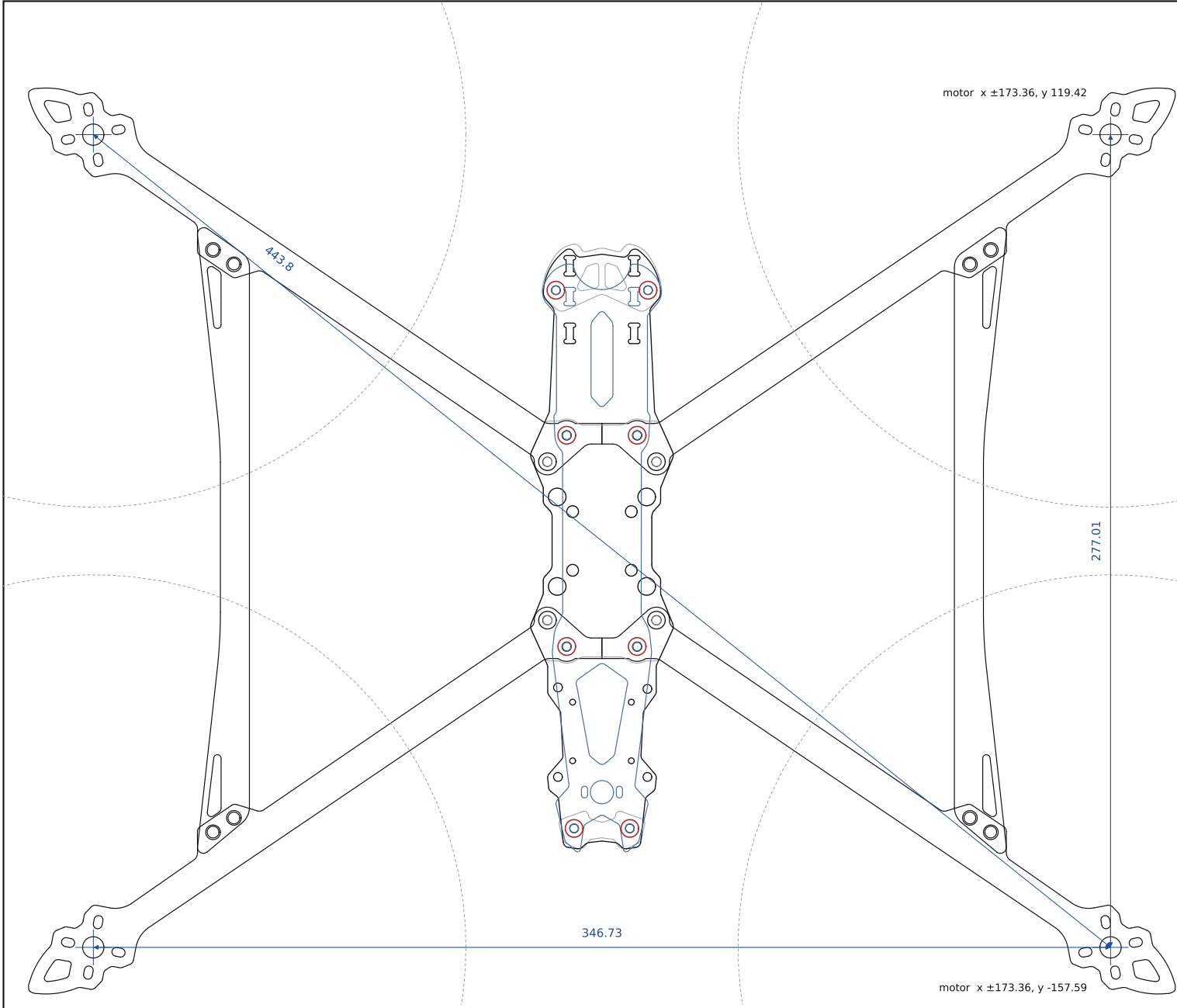




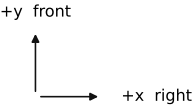
Parts

| Qty | Part (STEP)       | t mm | RJX manual      |
|-----|-------------------|------|-----------------|
| 1   | Top plate         | 2    | Top plate       |
| 1   | Lower plate       | 3    | Mid plate       |
| 1   | Belly plate       | 3    | Bottom plate    |
| 1   | Chin plate        | 2.25 | Front plate     |
| 1   | Tail plate        | 2.25 | Back plate      |
| 4   | Arm               | 7    | Arm             |
| 2   | Arm brace         | 2    | Arm bracer      |
| 2   | Camera plate      | 2.5  | Camera plate    |
| 8   | Standoff M3x35 Ø6 |      | Standoff spacer |

Stack-up: z in mm, 0 = underside of the lower plate

| Part              | z from | z to |
|-------------------|--------|------|
| top plate         | 38     | 40   |
| standoffs         | 3      | 38   |
| camera plates     | 0      | 40   |
| lower plate       | 0      | 3    |
| arm braces        | 0      | 2    |
| chin, tail plates | -2.25  | 0    |
| arms              | -7     | 0    |
| belly plate       | -10    | -7   |

Every part is a flat plate: extrude its DXF profile by its thickness, at its z.



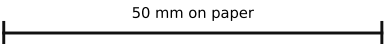
Seen from above, nose up.  
Dashed: 10" prop discs.  
Red: standoffs. Blue: top plate.

Frame overview

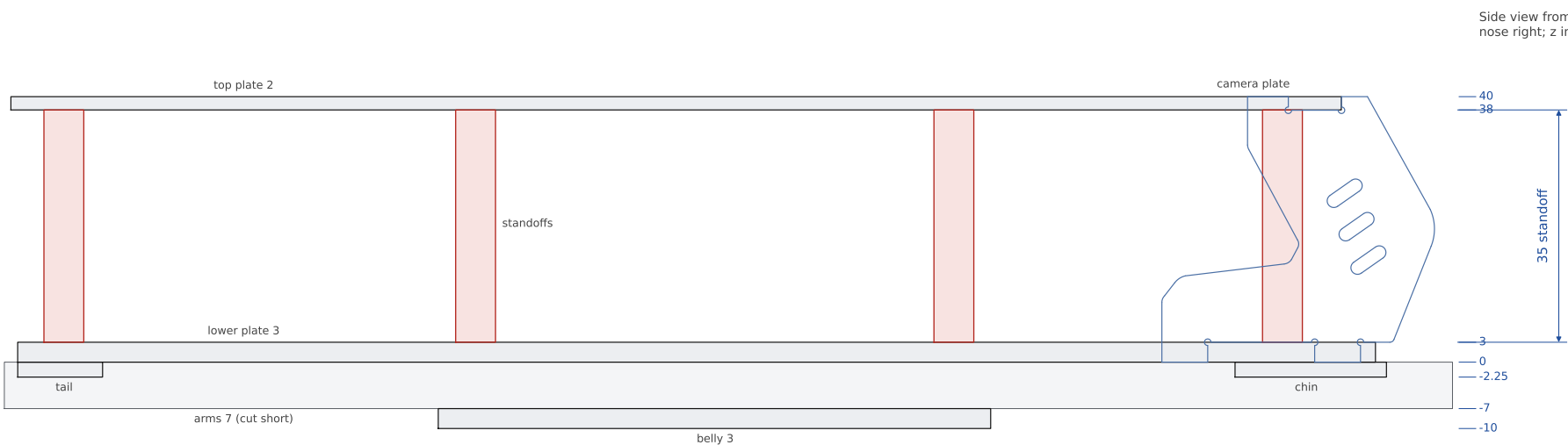
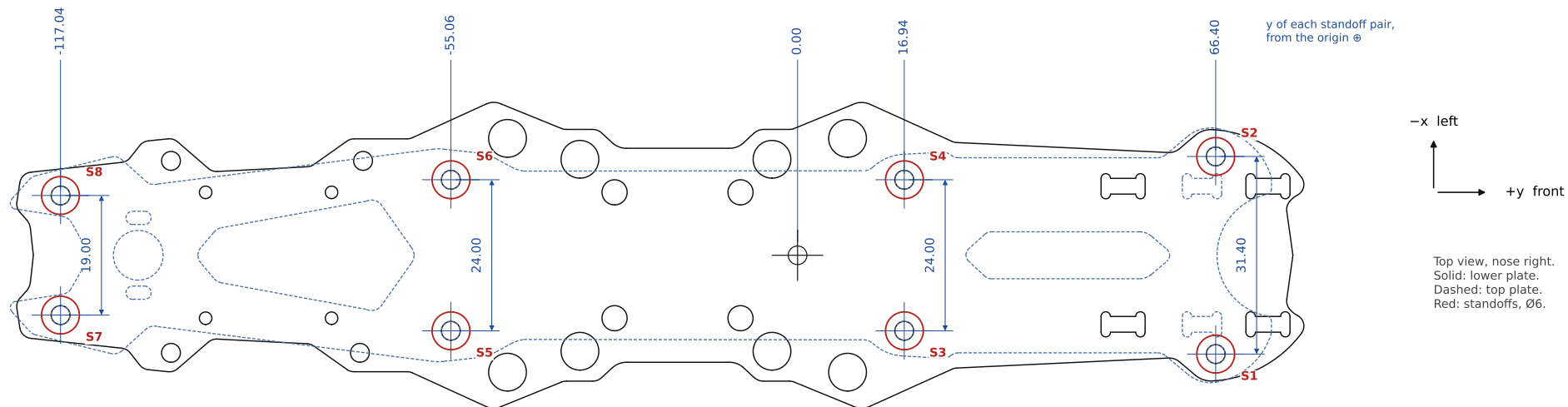
Scale 1:2

Motor centres 346.7 x 277.0 mm, 443.8 mm diagonal

Sheet 1 of 7



Mark4 V2 10" frame, from its STEP model. mm. Print at 100 % (actual size) and check the bar.

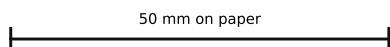


#### Standoff holes (lower plate; use these in every plate)

|    | x      | y     | Top plate | Bolts from below via |
|----|--------|-------|-----------|----------------------|
| S1 | 15.70  | 66.40 | same      | Chin                 |
| S2 | -15.70 | 66.40 | same      | Chin                 |
| S3 | 12.00  | 16.94 | y 16.97   | Arm, Belly           |
| S4 | -12.00 | 16.94 | y 16.97   | Arm, Belly           |

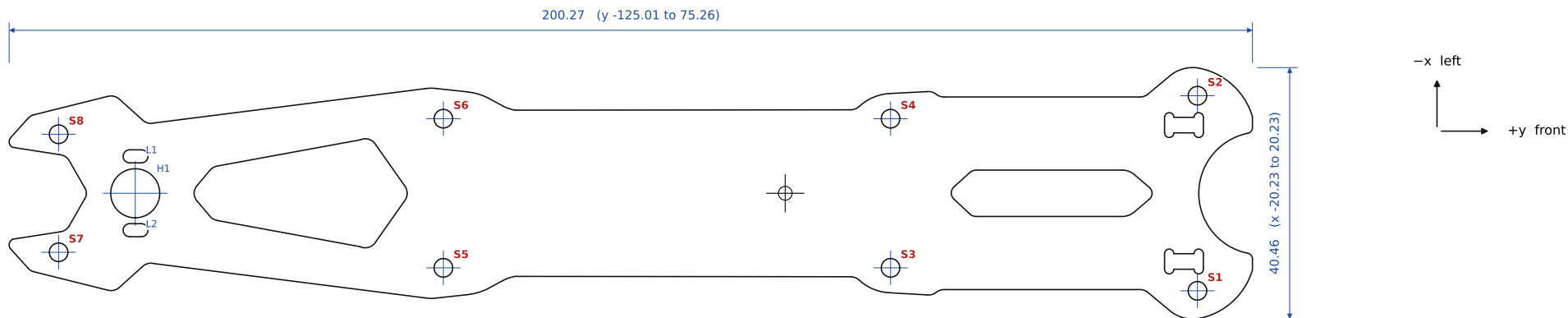
|    | x      | y       | Top plate | Bolts from below via |
|----|--------|---------|-----------|----------------------|
| S5 | 12.00  | -55.06  | same      | Arm, Belly           |
| S6 | -12.00 | -55.06  | same      | Arm, Belly           |
| S7 | 9.50   | -117.04 | same      | Tail                 |
| S8 | -9.50  | -117.04 | same      | Tail                 |

In the model every hole on a standoff axis is within 0.07 mm of it, and the top and lower plates within 0.03 mm of each other (CHECK.md).



Mark4 V2 10" frame, from its STEP model. mm. Print at 100 % (actual size) and check the bar.

|                                                        |              |
|--------------------------------------------------------|--------------|
| <b>Standoffs (struts) and stack-up</b>                 | Scale 1:1    |
| 8 x M3 x 35 mm standoffs, Ø6, lower plate to top plate | Sheet 2 of 7 |

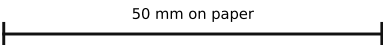


Hole centres from the model's origin @: x right, y forward. S1-S8: standoff holes, on the same axes in every plate. Thickness 2 mm.

4 cut-outs and the outline: take them from dxf/top\_plate.dxf.

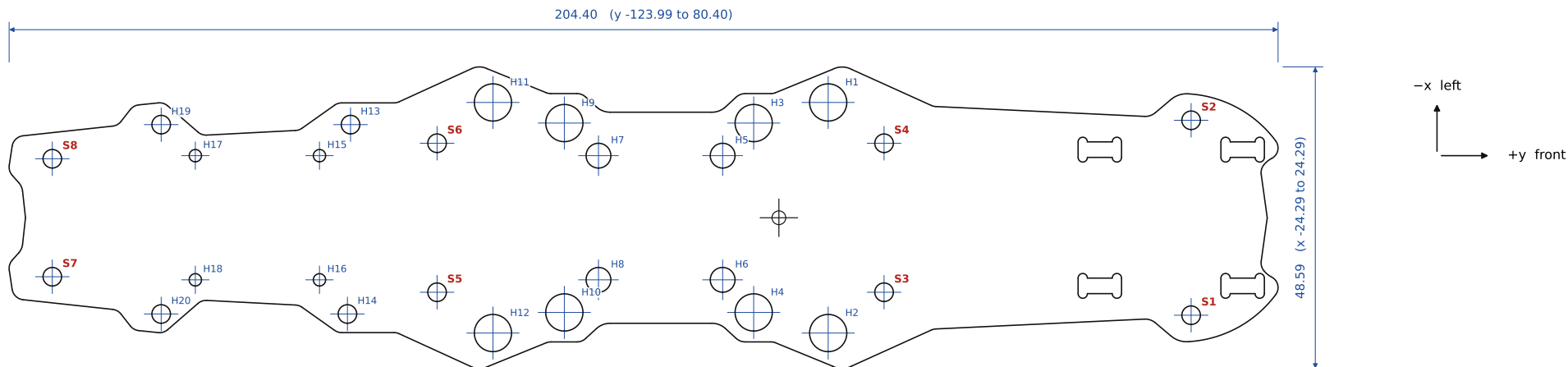
|    | x      | y       | Ø    |
|----|--------|---------|------|
| S2 | -15.70 | 66.40   | 3.00 |
| S1 | 15.70  | 66.40   | 3.00 |
| S4 | -12.00 | 16.97   | 3.00 |
| S3 | 12.00  | 16.97   | 3.00 |
| S6 | -12.00 | -55.06  | 3.00 |
| S5 | 12.00  | -55.06  | 3.00 |
| H1 | 0.00   | -104.70 | 7.90 |
| S8 | -9.50  | -117.04 | 3.00 |
| S7 | 9.50   | -117.04 | 3.00 |

| Slot | x1    | y1      | x2    | y2      | w    |
|------|-------|---------|-------|---------|------|
| L1   | -5.94 | -103.70 | -5.94 | -105.60 | 2.10 |
| L2   | 5.94  | -103.70 | 5.94  | -105.60 | 2.10 |



Mark4 V2 10" frame, from its STEP model. mm. Print at 100 % (actual size) and check the bar.

|                                    |              |
|------------------------------------|--------------|
| Top plate                          | Scale 1:1    |
| Qty 1. Carbon 2 mm. Battery on top | Sheet 3 of 7 |



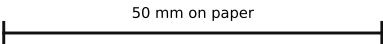
Hole centres from the model's origin @: x right, y forward. S1-S8: standoff holes, on the same axes in every plate. Thickness 3 mm.

4 cut-outs and the outline: take them from dxf/lower\_plate.dxf.

|    | x      | y     | Ø    |
|----|--------|-------|------|
| S2 | -15.70 | 66.40 | 3.00 |
| S1 | 15.70  | 66.40 | 3.00 |
| S4 | -12.00 | 16.94 | 3.00 |
| S3 | 12.00  | 16.94 | 3.00 |
| H1 | -18.57 | 7.94  | 6.00 |
| H2 | 18.57  | 7.94  | 6.00 |
| H3 | -15.25 | -4.06 | 6.00 |
| H4 | 15.25  | -4.06 | 6.00 |
| H5 | -10.00 | -9.06 | 4.00 |
| H6 | 10.00  | -9.06 | 4.00 |

|     | x      | y      | Ø    |
|-----|--------|--------|------|
| H7  | -10.00 | -29.06 | 4.00 |
| H8  | 10.00  | -29.06 | 4.00 |
| H9  | -15.25 | -34.56 | 6.00 |
| H10 | 15.25  | -34.56 | 6.00 |
| H11 | -18.57 | -46.06 | 6.00 |
| H12 | 18.57  | -46.06 | 6.00 |
| S6  | -12.00 | -55.06 | 3.00 |
| S5  | 12.00  | -55.06 | 3.00 |
| H13 | -15.00 | -69.00 | 3.00 |
| H14 | 15.50  | -69.50 | 3.00 |

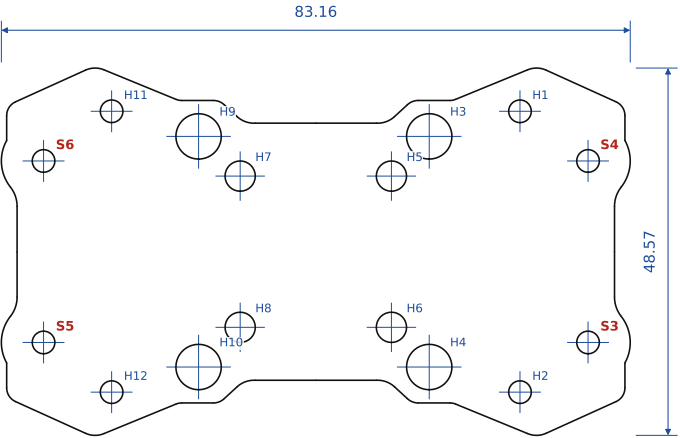
|     | x      | y       | Ø    |
|-----|--------|---------|------|
| H15 | -10.00 | -74.00  | 2.00 |
| H16 | 10.00  | -74.00  | 2.00 |
| H17 | -10.00 | -94.00  | 2.00 |
| H18 | 10.00  | -94.00  | 2.00 |
| H19 | -15.00 | -99.50  | 3.00 |
| H20 | 15.50  | -99.50  | 3.00 |
| S8  | -9.50  | -117.04 | 3.00 |
| S7  | 9.50   | -117.04 | 3.00 |



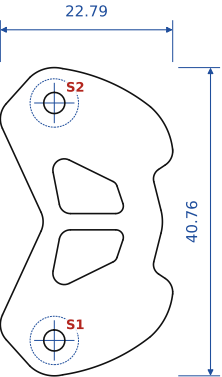
Mark4 V2 10" frame, from its STEP model. mm. Print at 100 % (actual size) and check the bar.

|                                              |              |
|----------------------------------------------|--------------|
| Lower plate (manual: mid plate)              | Scale 1:1    |
| Qty 1. Carbon 3 mm. Standoffs and the stacks | Sheet 4 of 7 |

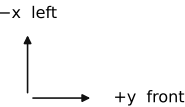
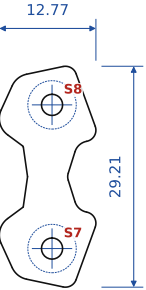
Belly plate



Chin plate



Tail plate



Belly plate

|    | x      | y     | Ø    |
|----|--------|-------|------|
| S4 | -12.00 | 16.90 | 3.00 |
| S3 | 12.00  | 16.90 | 3.00 |
| H1 | -18.57 | 7.90  | 3.00 |
| H2 | 18.57  | 7.90  | 3.00 |
| H3 | -15.25 | -4.10 | 6.00 |
| H4 | 15.25  | -4.10 | 6.00 |
| H5 | -10.00 | -9.10 | 4.00 |
| H6 | 10.00  | -9.10 | 4.00 |

|     | x      | y      | Ø    |
|-----|--------|--------|------|
| H7  | -10.00 | -29.10 | 4.00 |
| H8  | 10.00  | -29.10 | 4.00 |
| H9  | -15.25 | -34.60 | 6.00 |
| H10 | 15.25  | -34.60 | 6.00 |
| H11 | -18.57 | -46.10 | 3.00 |
| H12 | 18.57  | -46.10 | 3.00 |
| S6  | -12.00 | -55.10 | 3.00 |
| S5  | 12.00  | -55.10 | 3.00 |

Chin plate

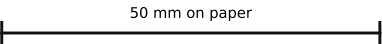
|    | x      | y     | Ø    |                       |
|----|--------|-------|------|-----------------------|
| S2 | -15.70 | 66.40 | 2.80 | csk Ø6.4 x 90°, under |
| S1 | 15.70  | 66.40 | 2.80 | csk Ø6.4 x 90°, under |

Tail plate

|    | x     | y       | Ø    |                       |
|----|-------|---------|------|-----------------------|
| S8 | -9.50 | -117.01 | 2.80 | csk Ø6.4 x 90°, under |
| S7 | 9.50  | -117.01 | 2.80 | csk Ø6.4 x 90°, under |

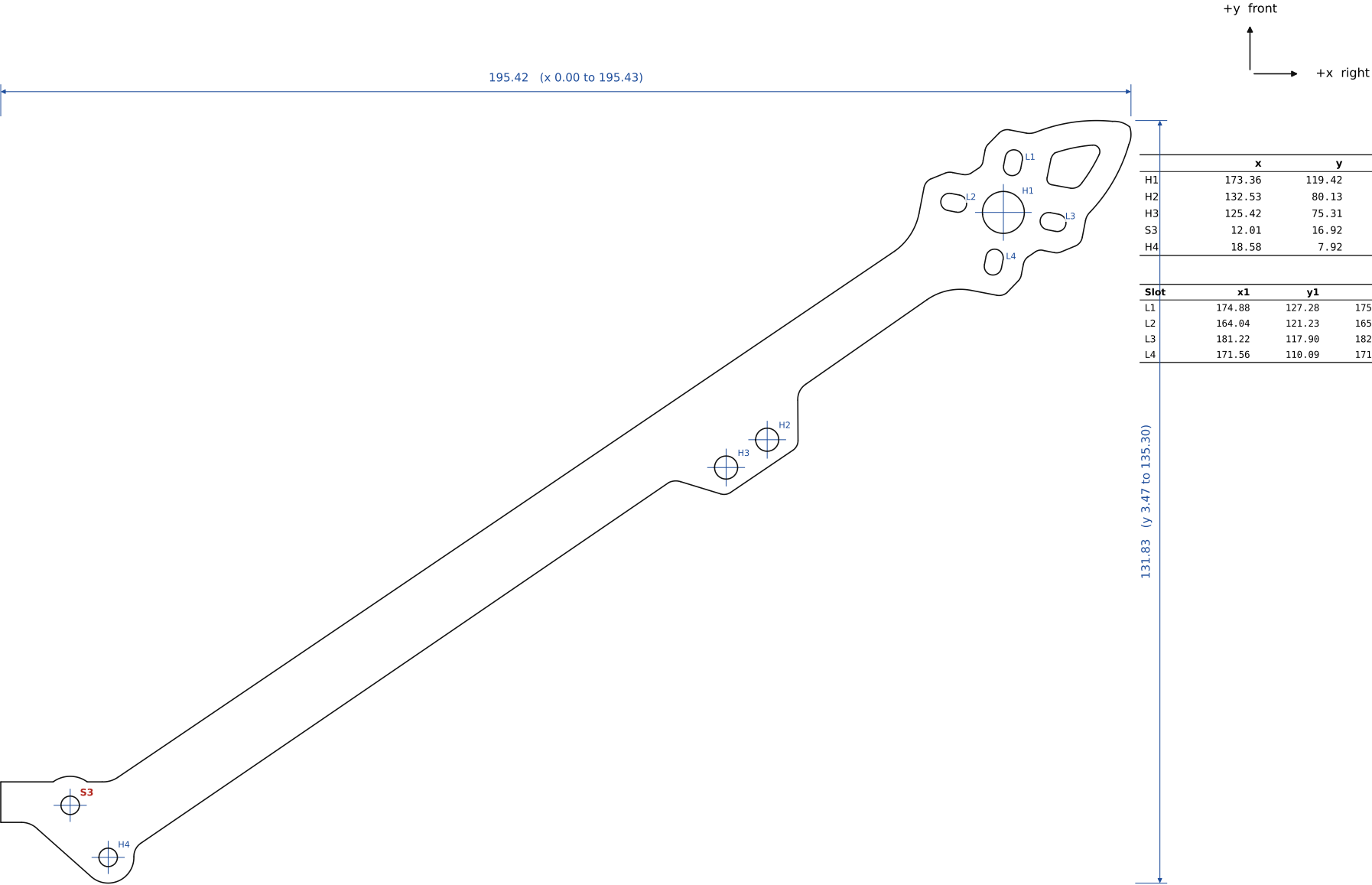
Chin and tail: Ø2.8 through, countersunk (dashed) on the underside, under standoffs S1, S2 and S7, S8.

Belly: under the arms; S3-S6 bolt up through it, the arms and the lower plate into the standoffs. Hole centres from the model's origin.



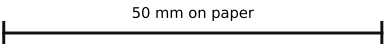
Mark4 V2 10" frame, from its STEP model. mm. Print at 100 % (actual size) and check the bar.

|                                                  |              |
|--------------------------------------------------|--------------|
| Belly, chin and tail plates                      | Scale 1:1    |
| 1 each. Belly carbon 3 mm; chin and tail 2.25 mm | Sheet 5 of 7 |



Motor centre H1: x 173.36, y 119.42, Ø7.25. The motor slots run from radius 8.00 to 9.50 about it: bolt circles Ø16.0 to Ø19.0 (16x16 and 19x19 motors).

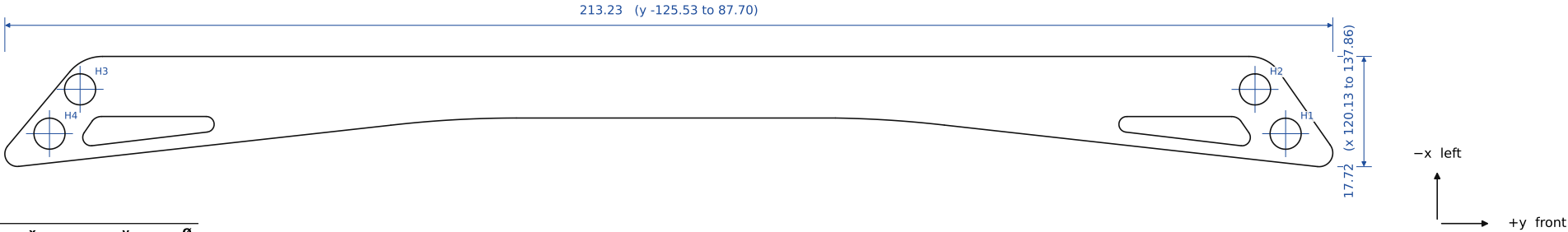
S3 (and S4-S6 on the mirrored arms) carries a standoff bolt. The outline and the 1 lightening cut-out come from dxf/arm.dxf.



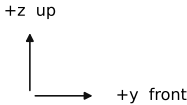
Mark4 V2 10" frame, from its STEP model. mm. Print at 100 % (actual size) and check the bar.

|                                                                     |              |
|---------------------------------------------------------------------|--------------|
| Arm                                                                 | Scale 1:1    |
| Qty 4. Carbon 7 mm. Front right drawn; the others are mirror images | Sheet 6 of 7 |

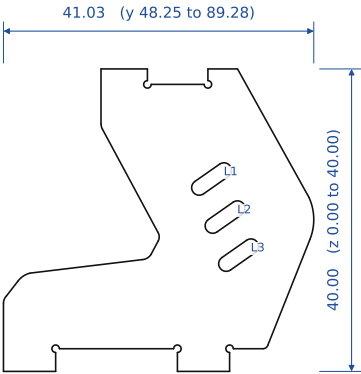
Arm brace (right), from above



|    | x      | y       | Ø    |
|----|--------|---------|------|
| H1 | 132.55 | 80.13   | 5.00 |
| H2 | 125.43 | 75.22   | 5.00 |
| H3 | 125.42 | -113.45 | 5.00 |
| H4 | 132.54 | -118.36 | 5.00 |



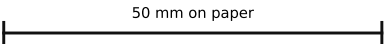
Camera plate (right), from the right



| Slot | y1    | z1    | y2    | z2    | w    |
|------|-------|-------|-------|-------|------|
| L1   | 74.13 | 24.30 | 77.41 | 26.59 | 2.00 |
| L2   | 75.92 | 19.26 | 79.19 | 21.56 | 2.00 |
| L3   | 77.70 | 14.23 | 80.98 | 16.52 | 2.00 |

Camera plates: z 0 to 40. Their tabs pass through the slots in the lower and top plates, flush with the outside faces. The right one's faces are at x 9.72 and 12.23.

Brace: on top of the arms, z 0 to 2; its holes line up with the arm's H2 and H3.



Mark4 V2 10" frame, from its STEP model. mm. Print at 100 % (actual size) and check the bar.

|                                               |              |
|-----------------------------------------------|--------------|
| Arm brace, camera plate                       | Scale 1:1    |
| Brace qty 2, 2 mm. Camera plate qty 2, 2.5 mm | Sheet 7 of 7 |